REGULAR INSULIN

High-alert drug

Only insulin for IV use (with analogues)

CHEMICAL NAME Human insulin (recombinant DNA): 51-amino-acid peptide, A-chain 21 and B-chain 30 residues linked by two disulfide bonds

CLASS Short-acting insulin (soluble human insulin)

TRADE NAMES Actrapid · Humulin R · Insuman Rapid · Novolin R

THERAPEUTIC USES / INDICATIONS

- Type 1 and type 2 diabetes (with meals).
- Diabetic ketoacidosis (DKA) and hyperosmolar hyperglycaemic state (IV infusion).
- Hyperkalaemia (with glucose).
- Perioperative and critical-illness glucose control (variable-rate IV insulin infusion).

ADMINISTRATION

- **SC** 30 min before meals (abdomen fastest absorption); rotate sites.
- **IV** infusion: 50 units in 50 mL 0.9% saline (1 unit/mL) via syringe pump.
- Use insulin syringes or pens only; write "units" in full (never "U").
- Clear solution — do not use if cloudy.

DOSAGES

- **SC:** individualised; usually 0.3–1.0 unit/kg/day total in basal-bolus regimens.
- **DKA:** fixed-rate IV infusion 0.1 unit/kg/h (+ continue basal insulin); when glucose <14 mmol/L (250 mg/dL) add 5–10% glucose (ADA/EASD 2024 also reduces rate to 0.05 unit/kg/h).
- **Hyperkalaemia:** 10 units in 50 mL 50% glucose (or 25 g glucose) IV over 15–30 min.
- **VRIII:** per local sliding scale.

PHARMACOKINETICS

- **Onset:** SC 30 min; IV immediate. **Peak:** SC 2–4 h. **Duration:** SC 6–8 h; IV t½ ~5 min.
- Metabolised in liver and kidney.
- Requirements fall in renal failure.

PHARMACODYNAMICS

- Binds **insulin receptors** → ↑ glucose uptake into muscle and fat (GLUT4), ↑ glycogen synthesis, ↓ hepatic glucose output, ↓ lipolysis and ketogenesis.
- Shifts **K^+ into cells**.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Injection-site reactions.
- Weight gain.
- Lipohypertrophy (repeated same-site injection).

ADVERSE EFFECTS (SERIOUS)

- Hypoglycaemia (sweating, tremor, confusion, seizures, coma).
- Hypokalaemia (IV, DKA treatment).
- Insulin errors (10-fold) → fatal hypoglycaemia.
- Allergy (rare).

CONTRAINDICATIONS & CAUTIONS

- Hypoglycaemia.
- Hypersensitivity to insulin or excipients.

KEY INTERACTIONS

- **β-blockers:** mask hypoglycaemia.
- **Steroids, thiazides, adrenaline:** ↑ glucose (↑ insulin need).
- **Alcohol, sulfonylureas, ACE inhibitors, quinolones:** ↑ hypoglycaemia risk.

NURSING MANAGEMENT

- Check capillary glucose before each dose; hourly glucose and K^+ (2-hourly) during IV infusion.
- Independent double-check of dose and device.
- Ensure food available after SC dose (30 min).
- Treat hypoglycaemia promptly (15–20 g fast-acting glucose; IV glucose or glucagon if unconscious).
- DKA: monitor ketones, K^+, fluid balance, GCS.

PATIENT EDUCATION

- Inject 30 minutes before meals; rotate sites within an area.
- Recognise and treat hypos; carry glucose tablets.
- Sick-day rules: never stop insulin when ill; check glucose and ketones often.
- Store unopened insulin in fridge; in-use at room temperature for 4 weeks.

KEY POINTS

- Short-acting insulin; SC onset 30 min.
- Hypoglycaemia and hypokalaemia.
- IV for DKA (0.1 unit/kg/h) and hyperkalaemia (10 units + glucose).
- "Units" in full; insulin syringes only.

SOURCES CHECKED **[586]** Actrapid 100 international units/ml, Solution for Injection in a vial (UK SmPC) **[587]** JBDS-IP DKA guideline (2023) **[448]** UKKA hyperkalaemia guideline (2023) **[588]** Umpierrez et al., Diabetologia 2024

Full references p. 432–457